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PROJECT AND TEAM INFORMATION

Project Title

EduBridge – A Role-Based Academic Collaboration and Management Platform

Student / Team Information

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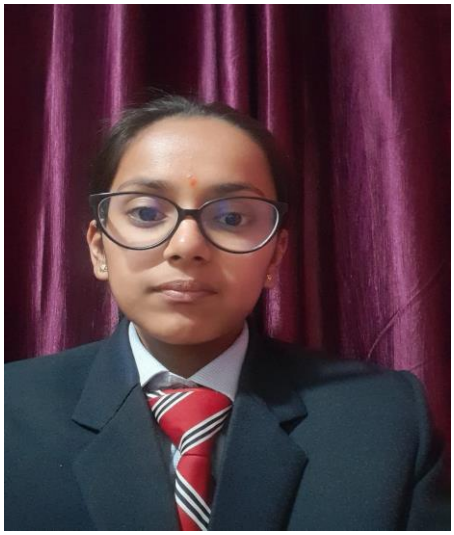
Team member 3	Guliyani, Gauri –230221455 guliyanigauri@gmail.com 

PROPOSAL DESCRIPTION

Motivation

In most colleges and universities, academic information is spread across different platforms. Students receive notices on WhatsApp, check emails for assignments, look at notice boards for updates, and teachers use Excel sheets to manage records. This scattered system creates many problems.

Students may miss important information because they have to check multiple places. Teachers spend extra time sharing the same message on different platforms. The administration finds it hard to track activities across departments. There is also no proper system to collect and analyze student feedback. Manual work wastes time and can lead to errors.

The main issues are:

- No single platform for academic communication
- Weak interaction between students and teachers
- No structured feedback system
- Time-consuming manual processes
- No proper access control

EduBridge aims to solve these problems by creating a single web-based platform where students, teachers, and management can communicate easily and securely. This will make academic life more organized and transparent for everyone.

State of the Art / Current solution

Right now, most educational institutions use one of these methods:

Traditional Methods:
Many colleges still use physical notice boards, paper feedback forms, and Excel sheets to keep records. This method takes a lot of time and can easily lead to mistakes.

WhatsApp and Email:
Some colleges use WhatsApp groups and email to share information. These are easy to use, but they are not made for academic purposes. They do not have proper organization, search options, or access control.

Large ERP Systems:
Some universities use big systems like SAP ERP or Oracle ERP. These systems are powerful but very expensive, complicated, and need special training to use.

Learning Management Systems:
Platforms like Moodle and Google Classroom mainly focus on study materials and assignments. They do not manage administrative communication, timetables, or feedback analysis properly.

The problem with these solutions is that some of them do very little (like traditional methods and WhatsApp), while others do too much and cost too much (like ERP systems). There is a need for a simple, affordable, web-based platform made especially for academic communication and collaboration.

Project Goals and Milestones

The main aim of this project is to build **EduBridge**, a central web platform that connects students, teachers, and management in one place. The goal is to make academic communication simple, safe, and efficient.

Primary Goals:

- Create one platform for all academic communication
- Build a secure login system with different access levels for students, teachers, and admin
- Make academic information clear and easy to access
- Provide real-time insights using analytics and reports

Secondary Goals:

- Improve communication between students and teachers
- Reduce manual work for the admin staff
- Help management make better decisions using data

Project Timeline (12 Weeks):

Weeks 1–2:
Plan the project, define requirements, design the database, and plan the user interface

Weeks 3–4:
Develop the login system and basic user management

Weeks 5–6:
Build the notice board and messaging features

Weeks 7–8:
Create the timetable management and feedback system

Weeks 9–10:
Develop the analytics dashboard and reporting features

Weeks 11–12:
Testing, fixing bugs, and completing documentation

Project Approach

We will build **EduBridge** using modern web technologies and follow proper industry standards. The system will have three main parts:

1. Frontend (What Users See)
We will use React to create the user interface. React is popular, easy to use, and helps build modern and responsive websites.
We will use:

- React for moving between pages
- Context API to manage user data like login status

The design will be simple, clean, and easy to use. We will focus more on usability.

2. Backend (Server Side)
We will use Node.js with Express.js to build the server.

- Node.js is fast and efficient
- Express helps us create REST APIs easily
- We will use JWT (JSON Web Tokens) for secure login
- Middleware will check user roles before giving access to features

All APIs will follow REST principles so they are simple and easy to maintain.

3. Database (Data Storage)
We will use PostgreSQL as our database because it is reliable, free, and works well with complex data.

- We will design tables carefully to avoid duplicate data
- We will use the **pg** library to connect Node.js with PostgreSQL
- SQL queries will be used to manage the data

4. Security Measures

- Passwords will be encrypted before storing in the database
- JWT tokens will expire after a certain time
- Role-based access control will limit what each user can see
- Input validation will be done on both frontend and backend to prevent harmful data

We will build the project step by step. Each feature will be tested before moving to the next one. This will help us find and fix problems early. We will also write clean and well-commented code so that it is easy to understand and update in the future.

Project Outcome / Deliverables

By the end of this project, we will build a fully working web application with these features:

Main Features

1. Login System:
Users can log in using email and password. Students, teachers, and management will have different dashboards.

2. Notice Board:
Teachers and management can post notices. Students can see notices based on their department or year.

3. Messaging:
Students and teachers can chat privately. Messages will be organized properly in an inbox.

4. Timetable:
Teachers can upload timetables. Students can view and download them.

5. Feedback:
Students can give ratings and comments to teachers. Teachers can see the overall feedback.

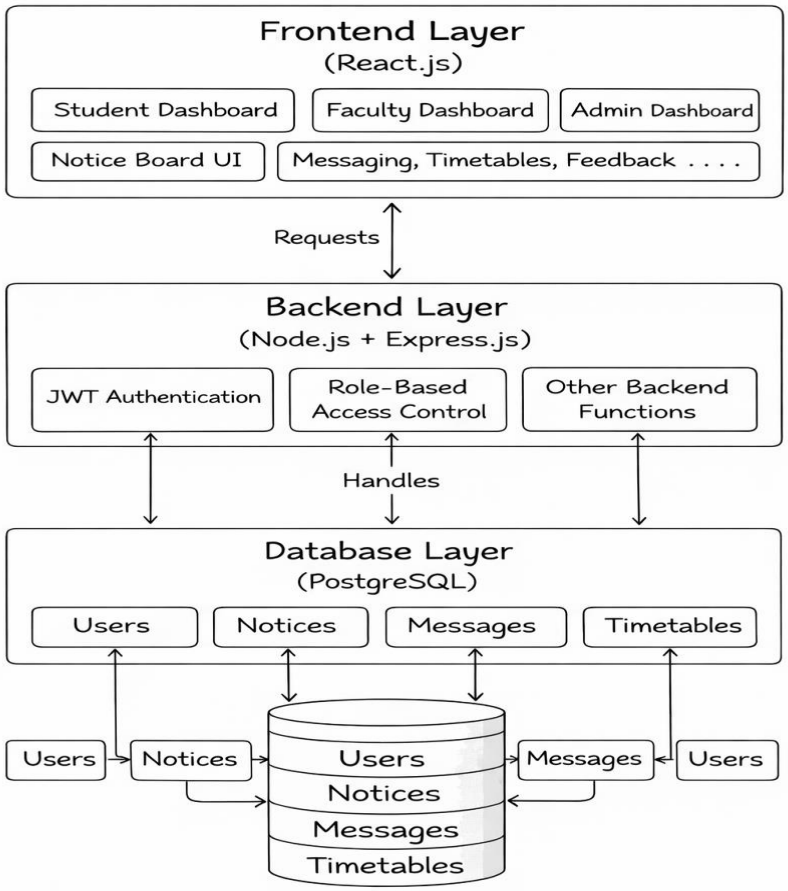
6. Analytics Dashboard:
Management can see simple reports like total students, total teachers, average feedback, and system usage.

What We Will Deliver

- Full source code with clear explanation
- Database design with all tables
- API documentation
- User guide to explain how to use the system
- Guide to deploy the application
- Final report explaining design decisions and challenges

The final application will run on a web browser. Users can open it on a computer or mobile phone without installing anything.

System Architecture (High Level Diagram) (3 pts)



Assumptions

- All users have access to the internet and a device such as a laptop, desktop, or smartphone.
- The system will be used within a college or university environment with stable internet connectivity.
- The management will be responsible for creating user accounts and assigning roles (student, faculty, and management).

- The initial version of the system will be used by a limited number of users during the testing phase.
- Timetable files uploaded by faculty will be in common formats like PDF, Excel, or image files.
- The system will focus only on web-based access (no mobile app in the first version).

References

[1] ERP – Graphic Era University (Cyborg-ERP)
The official ERP portal used by Graphic Era Deemed to be University where Students can access ERP, notices, results, exam portals, and other services.
Available at: https://erp.geu.ac.in/

[2] Moodle – Learning Management System
Moodle is a widely used open-source LMS platform for course management and online learning.

[3] Node.js Documentation
OpenJS Foundation. *Node.js Official Documentation.*
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[4] Express.js Documentation
Express.js Team. *Express.js Official Guide.*
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[5] PostgreSQL Documentation
PostgreSQL Global Development Group. *PostgreSQL Official Documentation.*
Available at: https://www.postgresql.org/docs/

[6] REST API Design Best Practices
RESTfulAPI.net. *REST API Design Guide.*
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